

T-Test

The **t-test** is a statistical hypothesis testing technique developed by **William Sealy Gosset** under the pseudonym "Student". It is used to determine whether there is a significant difference between means when:

- The sample size is small (generally ($n < 30$))
- The population standard deviation is unknown
- Data are approximately normally distributed

T-tests are widely used in research, education, medicine, business, and social sciences to test assumptions and compare groups.

Objectives of T-Test

The t-test is used to:

1. Compare a sample mean with a known population mean.
2. Compare the means of two independent groups.
3. Compare means of the same group measured at different times.
4. Determine whether observed differences are statistically significant.
5. Test research hypotheses.

Assumptions of T-Test

Before applying a t-test, the following assumptions should be met:

1. Data are numerical.
2. Samples are randomly selected.
3. Data are approximately normally distributed.
4. Observations are independent.
5. Population standard deviation is unknown.

Types of T-Test

The most common t-tests are:

1. **Single Sample T-Test**
 - Mean and standard deviation given.
 - Raw data given.
2. **Independent Sample T-Test**
 - Comparison of two unrelated groups.

3. Paired Sample (Dependent Sample) T-Test

- Comparison of before-after or matched observations.

One-Tailed and Two-Tailed T-Test

One-Tailed Test

Used when the researcher expects a difference in a specific direction.

Right-Tailed Test

$$H_0: \mu = \mu_0$$

$$H_1: \mu > \mu_0$$

Left-Tailed Test

$$H_0: \mu = \mu_0$$

$$H_1: \mu < \mu_0$$

Two-Tailed Test

Used when the researcher expects a difference but does not know the direction.

$$H_0: \mu = \mu_0$$

$$H_1: \mu \neq \mu_0$$

General Procedure of Hypothesis Testing Using T-Test

Step 1: State Hypotheses

Null Hypothesis:

$$H_0$$

Alternative Hypothesis:

$$H_1$$

Step 2: Select Significance Level

Generally:

$$\alpha = 0.05$$

or

$$\alpha = 0.01$$

Step 3: Calculate Test Statistic

Use the appropriate t-test formula.

Step 4: Determine Degrees of Freedom

Depends on test type.

Step 5: Find Critical Value

Use t-distribution table.

Step 6: Decision

If:

$$|t_{\text{calculated}}| > t_{\text{critical}}$$

Reject (H_0) or **Accept Alternative Hypothesis(H_1)**

Otherwise:

Fail to Reject(H_0) **Accept H_0** **Reject Alternative Hypothesis(H_1)**

Single Sample T-Test (Mean and Standard Deviation Given)

Purpose

To determine whether a sample mean differs significantly from a known population mean.

Formula

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

Where:

Symbol	Meaning
$(\bar{X})$	Sample Mean
(μ)	Population Mean
(s)	Sample Standard Deviation
(n)	Sample Size

Example 1

Problem: A school claims that the average mathematics score is 70.

A sample of 25 students gives:

$$\begin{aligned}\bar{X} &= 75 \\ s &= 10\end{aligned}$$

Test the claim at 5% significance level.

Step 1: Hypothesis

$$\begin{aligned}H_0: \mu &= 70 \\ H_1: \mu &\neq 70\end{aligned}$$

Step 2: Given Data

$$\bar{X} = 75$$

$$\mu = 70$$

$$s = 10$$

$$n = 25$$

Step 3: Calculate t-value

Formula:

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

Substitute values:

$$t = \frac{75 - 70}{10/\sqrt{25}}$$

$$t = \frac{5}{10/5}$$

$$t = \frac{5}{2}$$

$$t = 2.5$$

Step 4: Degree of Freedom

$$df = n - 1$$

$$df = 25 - 1$$

$$df = 24$$

Step 5: Critical Value

For:

$$\alpha = 0.05$$

Two-tailed:

$$t_{0.025,24} = 2.064$$

Step 6: Decision

$$2.5 > 2.064$$

Reject(H_0)

Conclusion

The sample mean differs significantly from 70.

Example 2

Given:

$$\bar{X} = 43$$

$$\mu = 40$$

$$s = 6$$

$$n = 16$$

Solution

$$t = \frac{43 - 40}{6/\sqrt{16}}$$

$$t = \frac{3}{6/4}$$

$$t = \frac{3}{1.5}$$

$$t = 2.0$$

$$df = 15$$

Critical value:

$$2.131$$

Since

$$2.0 < 2.131$$

Fail to reject (H_0)

Practice Exercises

Exercise 1

$$\bar{X} = 84, \mu = 80, s = 8, n = 25$$

Exercise 2

$$\bar{X} = 46, \mu = 50, s = 7, n = 20$$

Single Sample T-Test (Raw Data Given)

When individual observations are provided, mean and standard deviation must first be calculated.

Formula for Mean

$$\bar{X} = \frac{\sum X}{n}$$

Formula for Standard Deviation

$$s = \sqrt{\frac{\sum (X - \bar{X})^2}{n - 1}}$$

Formula for T-Test

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

Example 1

Data

12,14,13,15,16

Population mean:

$$\mu = 12$$

Step 1: Mean

$$\bar{X} = \frac{12 + 14 + 13 + 15 + 16}{5}$$

$$\bar{X} = 14$$

Step 2: Standard Deviation

X	X-14	(X-14) ²
12	-2	4
14	0	0
13	-1	1
15	1	1
16	2	4

$$\sum (X - \bar{X})^2 = 10$$

$$s = \sqrt{\frac{10}{4}}$$

$$s = 1.581$$

Step 3: Calculate t-value

$$t = \frac{14 - 12}{1.581/\sqrt{5}}$$

$$t = \frac{2}{0.707}$$

$$t = 2.83$$

Step 4

$$df = n - 1 = 5 - 1 = 4$$

Step 5

Critical value: 2.776

Step 6

$$[2.83 > 2.776]$$

Reject(H_0)

Example 2

Data:

[20,22,24,21,23]

Test against

$$\mu = 20$$

Solution

$$\bar{X} = 22$$

$$s = 1.581$$

$$t = \frac{22 - 20}{1.581/\sqrt{5}}$$

$$t = 2.83$$

Practice Exercises

Exercise 1

8,10,12,11,9

Test against:

$$\mu = 10$$

Exercise 2

18,20,22,24,26

Test against:

$$\mu = 20$$

Independent Sample T-Test (Double Mean T-Test)

Used to compare means of two **unrelated** groups.

Examples

- Male vs Female
- School A vs School B

- Urban vs Rural Students

Hypotheses

$$[H_0: \mu_1 = \mu_2]$$

$$[H_1: \mu_1 \neq \mu_2]$$

Pooled Variance Formula

$$\left[S_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \right]$$

T-Test Formula

$$\left[t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \right]$$

Example 1

Group A:

$$[n_1 = 12]$$

$$[\bar{X}_1 = 34]$$

$$[s_1 = 7]$$

Group B:

$$[n_2 = 10]$$

$$[\bar{X}_2 = 28]$$

$$[s_2 = 7]$$

Step 1

$$[H_0: \mu_1 = \mu_2]$$

$$[H_1: \mu_1 \neq \mu_2]$$

Step 2: Pooled Variance

$$\left[S_p^2 = \frac{11(49) + 9(49)}{20} \right]$$

$$[S_p^2 = 49]$$

Step 3: Calculate t-value

$$\left[t = \frac{34 - 28}{\sqrt{49 \left(\frac{1}{12} + \frac{1}{10} \right)}} \right]$$

$$\left[t = \frac{6}{\sqrt{8.98}} \right]$$

$$\left[t = \frac{6}{2.996} \right]$$

$$[t = 2.00]$$

Step 4: Degrees of Freedom

$$[df = n_1 + n_2 - 2]$$

$$[df = 20]$$

Step 5

Critical value:

$$[2.086]$$

Step 6

$$[2.00 < 2.086]$$

Fail to reject (H_0)

Example 2

Group A:

$$[n = 15, \bar{X} = 70, s = 8]$$

Group B:

$$[n = 15, \bar{X} = 65, s = 7]$$

Solution

$$[S_p^2 = 56.5]$$

$$t = 1.82$$

$$[df = 28]$$

Fail to reject (H_0)

Practice Exercises

Exercise 1

Group A:

$$n = 10$$

$$\bar{X} = 40$$

$$s = 5$$

Group B:

$$n = 12$$

$$\bar{X} = 35$$

$$s = 6$$

Exercise 2

Group A:

$$[n = 15, \bar{X} = 78, s = 8]$$

Group B:

$$[n = 15, \bar{X} = 72, s = 7]$$

Paired Sample T-Test (Dependent Sample T-Test)

Used when observations are paired.

Examples

- Before and After Training
- Before and After Treatment
- Pre-Test and Post-Test Scores

Formula

Difference:

$$[d = X_2 - X_1]$$

Mean Difference:

$$\left[\bar{d} = \frac{\sum d}{n} \right]$$

Standard Deviation:

$$\left[s_d = \sqrt{\frac{\sum (d - \bar{d})^2}{n - 1}} \right]$$

T Statistic:

$$\left[t = \frac{\bar{d}}{s_d / \sqrt{n}} \right]$$

Example 1

Before	After	d
40	45	5
42	47	5
38	42	4

45	50	5
41	46	5

Step 1

$$\left[\sum d = 24 \right]$$

$$[n = 5]$$

Step 2

$$\left[\bar{d} = \frac{24}{5} \right]$$

$$[\bar{d} = 4.8]$$

Step 3

d	d-4.8	(d-4.8) ²
5	0.2	0.04
5	0.2	0.04
4	-0.8	0.64
5	0.2	0.04
5	0.2	0.04

$$\left[\sum (d - \bar{d})^2 = 0.80 \right]$$

$$\left[s_d = \sqrt{\frac{0.80}{4}} \right]$$

$$[s_d = 0.447]$$

Step 4: Calculate t-value

$$\left[t = \frac{4.8}{0.447/\sqrt{5}} \right]$$

$$\left[t = \frac{4.8}{0.20} \right]$$

$$[t = 24]$$

Step 5

$$df = 4$$

Criticalvalue: [2.776]

Step 6

$$[24 > 2.776]$$

Reject (H_0)

Example 2

Differences:

$$[2,1,3,1,2]$$

$$[\bar{d} = 1.8]$$

$$[s_d = 0.837]$$

$$\left[t = \frac{1.8}{0.837/\sqrt{5}} \right]$$

$$[t = 4.81]$$

Practice Exercises

Exercise 1

Before	After
50	55
48	52
52	58
47	50
49	54

Exercise 2

Before	After
20	22
21	23

22	25
24	26
23	24

Test	Formula
Single Sample	$\left(t = \frac{\bar{X} - \mu}{s/\sqrt{n}} \right)$
Independent Sample	$\left(t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{S_p^2(1/n_1 + 1/n_2)}} \right)$
Paired Sample	$\left(t = \frac{\bar{d}}{s_d/\sqrt{n}} \right)$
Mean	$\left(\bar{X} = \frac{\sum X}{n} \right)$
Standard Deviation	$\left(s = \sqrt{\frac{\sum (X - \bar{X})^2}{n - 1}} \right)$
Degree of Freedom (Single)	$(n - 1)$
Degree of Freedom (Independent)	$(n_1 + n_2 - 2)$
Degree of Freedom (Paired)	$(n - 1)$

Example: Independent Sample T-Test

Problem

The examination scores of two groups of students are given below:

Group A

$$[n_1 = 12]$$

$$[\bar{X}_1 = 34]$$

$$[s_1 = 7]$$

Group B

$$[n_2 = 10]$$

$$[\bar{X}_2 = 28]$$

$$[s_2 = 7]$$

Test whether there is a significant difference between the means at 5% significance level.

Step 1: State Hypotheses

Null Hypothesis:

$$[H_0: \mu_1 = \mu_2]$$

Alternative Hypothesis:

$$[H_1: \mu_1 \neq \mu_2]$$

Step 2: Write Given Data

$$[n_1 = 12]$$

$$[\bar{X}_1 = 34]$$

$$[s_1 = 7]$$

$$[n_2 = 10]$$

$$[\bar{X}_2 = 28]$$

$$[s_2 = 7]$$

Step 3: Calculate Pooled Variance

Formula:

$$\left[S_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \right]$$

Substitute values:

$$\left[S_p^2 = \frac{(12 - 1)(7)^2 + (10 - 1)(7)^2}{12 + 10 - 2} \right]$$

$$\left[S_p^2 = \frac{11(49) + 9(49)}{20} \right]$$

$$\left[S_p^2 = \frac{539 + 441}{20} \right]$$

$$\left[S_p^2 = \frac{980}{20} \right]$$

$$[S_p^2 = 49]$$

Step 4: Calculate t-value

Formula:

$$\left[t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \right]$$

Substitute values:

$$\left[t = \frac{34 - 28}{\sqrt{49 \left(\frac{1}{12} + \frac{1}{10} \right)}} \right]$$

Calculate numerator: $[34 - 28 = 6]$

Therefore,

$$\left[t = \frac{6}{\sqrt{49 \left(\frac{1}{12} + \frac{1}{10} \right)}} \right]$$

Calculate Fraction

$$\left[\frac{1}{12} + \frac{1}{10} \right]$$

LCM of 12 and 10: $[LCM = 60]$

$$\left[\frac{1}{12} \frac{5}{60} \right]$$

$$\left[\frac{1}{10} \frac{6}{60} \right]$$

$$\left[\frac{5}{60} + \frac{6}{60} \frac{11}{60} \right]$$

$$\left[\frac{11}{60} 0.1833 \right]$$

Multiply by 49

$$[49 \times 0.18338.9817]$$

Take Square Root

$$[\sqrt{8.98172.9969}]$$

Final t-value

$$\left[t = \frac{6}{2.9969} \right] [t = 2.002] [t \approx 2.00]$$

Step 5: Degree of Freedom

Formula:

$$[df = n_1 + n_2 - 2]$$

Substitute values:

$$df = 12 + 10 - 2$$

$$[df = 20]$$

Step 6: Critical Value

At

$$\alpha = 0.05$$

(two-tailed)

and

$$[df = 20]$$

From the t-table:

$$[t_{critical} = 2.086]$$

Step 7: Decision

Calculated value:

$$[t = 2.00]$$

Critical value:

$$[t = 2.086]$$

Since

$$[2.00 < 2.086]$$

Fail to reject (H_0).

Step 8: Conclusion

There is no significant difference between the mean scores of Group A and Group B at the 5% significance level.